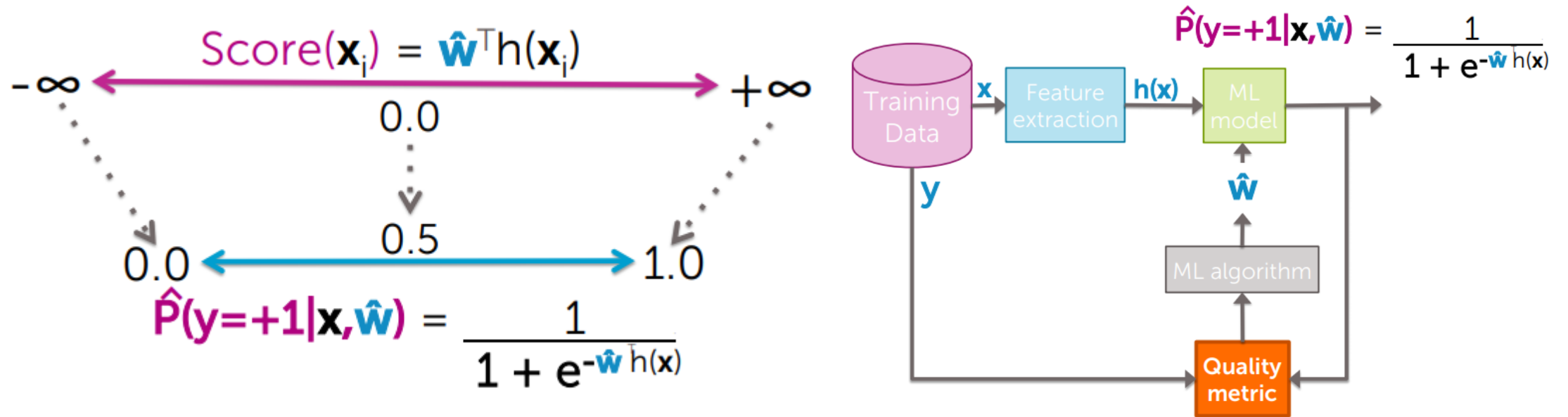


Clasificadores lineales – función de costo y regularización

Jhon Quiza

¿Cómo aprende el clasificador lineal?



Métrica de calidad – Función de verosimilitud

x[1] = #awesome	x[2] = #awful	y = sentiment
0	2	-1
3	3	-1
2	4	-1
0	3	-1
0	1	-1

$$P(y=+1|\mathbf{x}_i, \mathbf{w}) = 0.0$$

x[1] = #awesome	x[2] = #awful	y = sentiment
2	1	+1
4	1	+1
1	1	+1
2	1	+1

$$P(y=+1|\mathbf{x}_i, \mathbf{w}) = 1.0$$

Negative data points

$$P(y=+1|\mathbf{x}_i, \mathbf{w}) = 0.0$$

Positive data points

$$P(y=+1|\mathbf{x}_i, \mathbf{w}) = 1.0$$

No $\hat{\mathbf{w}}$ achieves perfect predictions (usually)

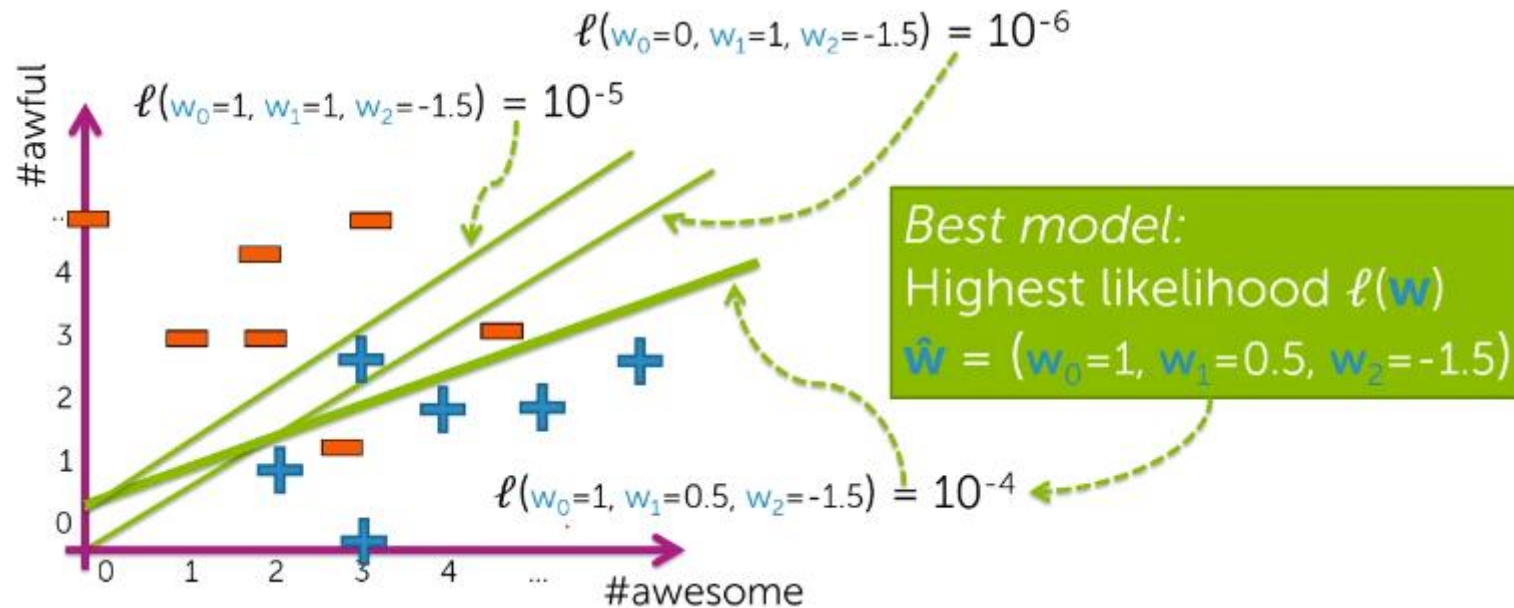
Estimación de máxima verosimilitud - MLE

Data point	x[1]	x[2]	y	Choose w to maximize
$\mathbf{x}_1, y_1$	2	1	+1	$P(y=+1 \mathbf{x}[1]=2, \mathbf{x}[2]=1, \mathbf{w})$
$\mathbf{x}_2, y_2$	0	2	-1	$P(y=-1 \mathbf{x}[1]=0, \mathbf{x}[2]=2, \mathbf{w})$
$\mathbf{x}_3, y_3$	3	3	-1	$P(y=-1 \mathbf{x}[1]=3, \mathbf{x}[2]=3, \mathbf{w})$
$\mathbf{x}_4, y_4$	4	1	+1	$P(y=+1 \mathbf{x}[1]=4, \mathbf{x}[2]=1, \mathbf{w})$

$$\begin{aligned} \ell(\mathbf{w}) = & \underbrace{P(y=+1|\mathbf{x}[1]=2, \mathbf{x}[2]=1, \mathbf{w})}_{P(y_1|\mathbf{x}_1, \mathbf{w})} \underbrace{P(y=-1|\mathbf{x}[1]=0, \mathbf{x}[2]=2, \mathbf{w})}_{P(y_2|\mathbf{x}_2, \mathbf{w})} \underbrace{P(y=-1|\mathbf{x}[1]=3, \mathbf{x}[2]=3, \mathbf{w})}_{P(y_3|\mathbf{x}_3, \mathbf{w})} \underbrace{P(y=+1|\mathbf{x}[1]=4, \mathbf{x}[2]=1, \mathbf{w})}_{P(y_4|\mathbf{x}_4, \mathbf{w})} \\ & \underbrace{\hspace{15em}}_{\prod_{i=1}^N P(y_i | \mathbf{x}_i, \mathbf{w})} \end{aligned}$$

Función de verosimilitud

Likelihood $\ell(\mathbf{w})$: Measures quality of fit for model with coefficients $\mathbf{w}$



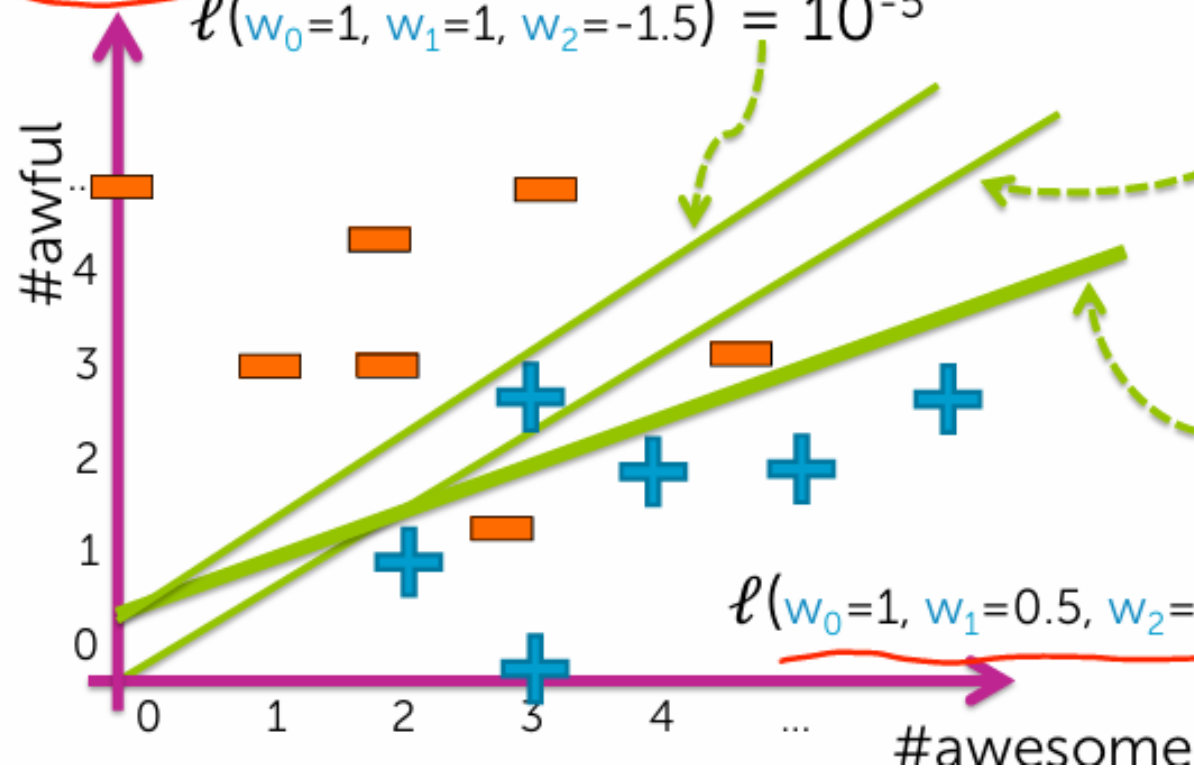
Find "best" classifier

Maximize likelihood over all possible w_0, w_1, w_2

$$\ell(\mathbf{w}) = \prod_{i=1}^N P(y_i \mid \mathbf{x}_i, \mathbf{w})$$

$$\ell(w_0=0, w_1=1, w_2=-1.5) = 10^{-6}$$

$$\ell(w_0=1, w_1=1, w_2=-1.5) = 10^{-5}$$



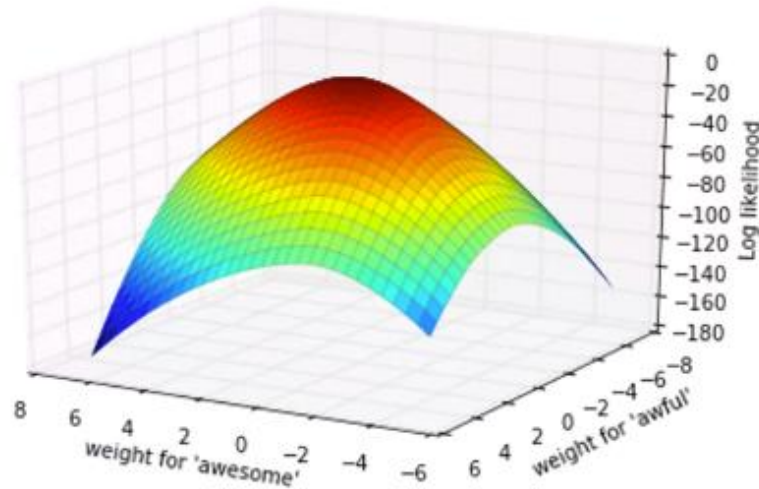
Best model:

Highest likelihood $\ell(\mathbf{w})$

$$\hat{\mathbf{w}} = (w_0=1, w_1=0.5, w_2=-1.5)$$

optimize with
gradient ascent

Maximización de la función de verosimilitud – Gradiente ascendiente

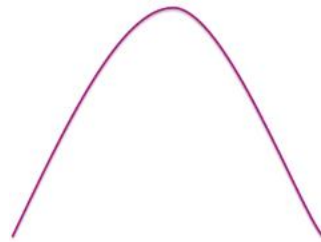


No closed-form solution → use gradient ascent

Maximize function over all possible w_0, w_1, w_2

$$\max_{w_0, w_1, w_2} \prod_{i=1}^N P(y_i | \mathbf{x}_i, \mathbf{w})$$

$\ell(w_0, w_1, w_2)$ is a function of 3 variables



Algorithm:

while not converged
 $w^{(t+1)} \leftarrow w^{(t)} + \eta \frac{d\ell}{dw} \Big|_{w^{(t)}}$

Gradiente ascendiente para regresión logística

$$\ell(\mathbf{w}) = \prod_{i=1}^N P(y_i \mid \mathbf{x}_i, \mathbf{w}) \Rightarrow \ell\ell(\mathbf{w}) = \ln \prod_{i=1}^N P(y_i \mid \mathbf{x}_i, \mathbf{w}) \Rightarrow \ell\ell(\mathbf{w}) = \sum_{i=1}^N \ln P(y_i \mid \mathbf{x}_i, \mathbf{w})$$

$$= \sum_{i=1}^N [\mathbb{1}[y_i = +1] \ln P(y = +1 \mid \mathbf{x}_i, \mathbf{w}) + \mathbb{1}[y_i = -1] \ln P(y = -1 \mid \mathbf{x}_i, \mathbf{w})]$$

Función indicadora:

$$\mathbb{1}[y_i = +1] = \begin{cases} 1, & \text{si } y_i = +1 \\ 0, & \text{si } y_i = -1 \end{cases}$$

Computing derivative

$$\frac{\partial \ell(\mathbf{w}^{(t)})}{\partial \mathbf{w}_j} = \sum_{i=1}^N h_j(\mathbf{x}_i) \left(\mathbb{1}[y_i = +1] - P(y = +1 | \mathbf{x}_i, \mathbf{w}^{(t)}) \right)$$

$\mathbf{w}^{(t)}$:

$w_0^{(t)}$	0
<u>$w_1^{(t)}$</u>	<u>1</u>
$w_2^{(t)}$	-2

$$\frac{\partial \ell}{\partial w_1}$$

$h_j(\mathbf{x}) = \# \text{ awesome}$

$x[1]$	$x[2]$	y	$P(y=+1 \mathbf{x}_i, \mathbf{w})$	Contribution to derivative for w_1
<u>2</u>	1	<u>+1</u>	<u>0.5</u>	$2(1 - 0.5) = 1$
<u>0</u>	2	<u>-1</u>	<u>0.02</u>	$0(0 - 0.02) = 0$
<u>3</u>	<u>3</u>	<u>-1</u>	<u>0.05</u>	$3(0 - 0.05) = -0.15$
<u>4</u>	1	<u>+1</u>	<u>0.88</u>	$4(1 - 0.88) = 0.48$

Total derivative:

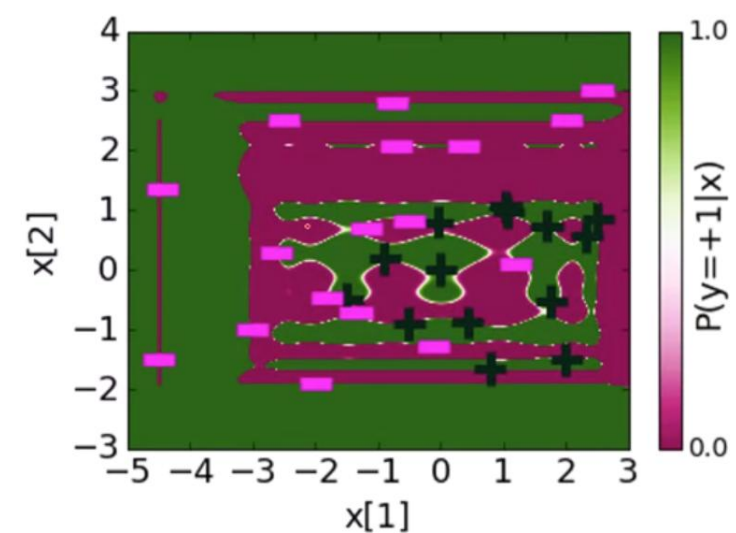
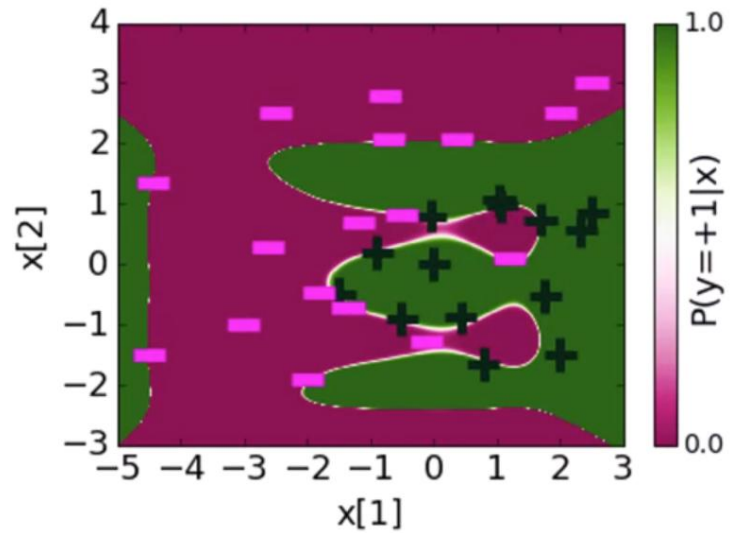
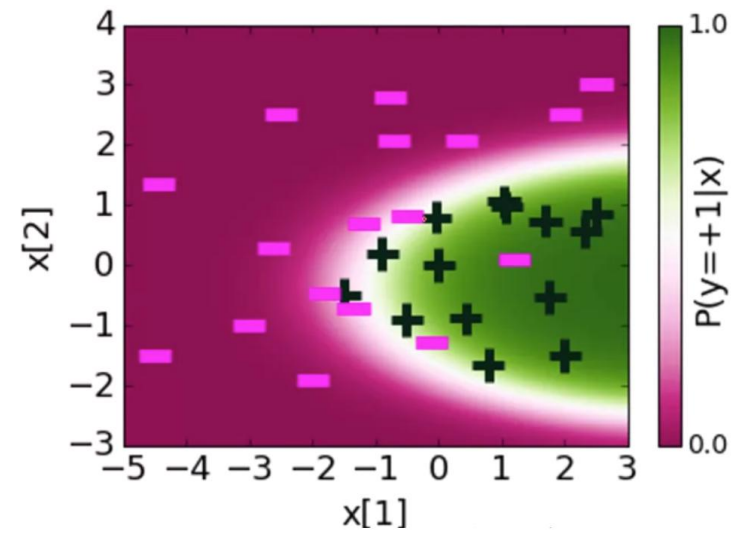
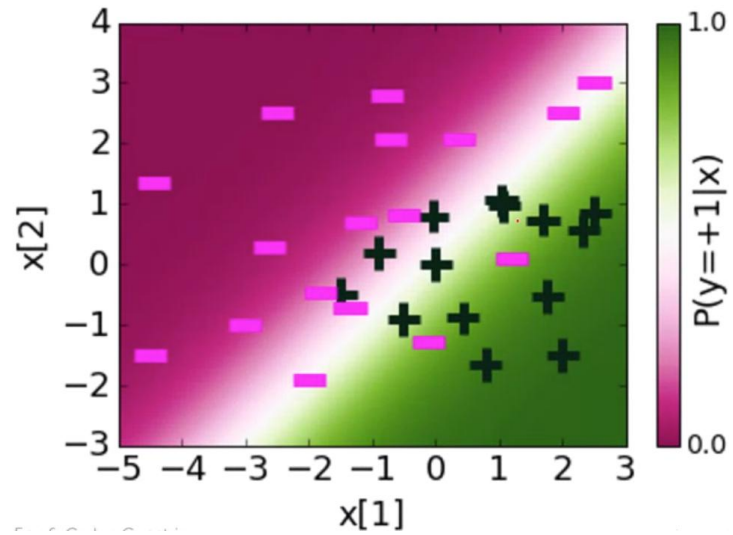
$$\frac{\partial \ell(\mathbf{w}^{(t)})}{\partial w_1} = 1 + 0 - 0.15 + 0.48 = 1.33$$

$$w_1^{(t+1)} = w_1^{(t)} + \eta \frac{\partial \ell(\mathbf{w}^{(t)})}{\partial w_1} \quad | \quad \eta = 0.1$$

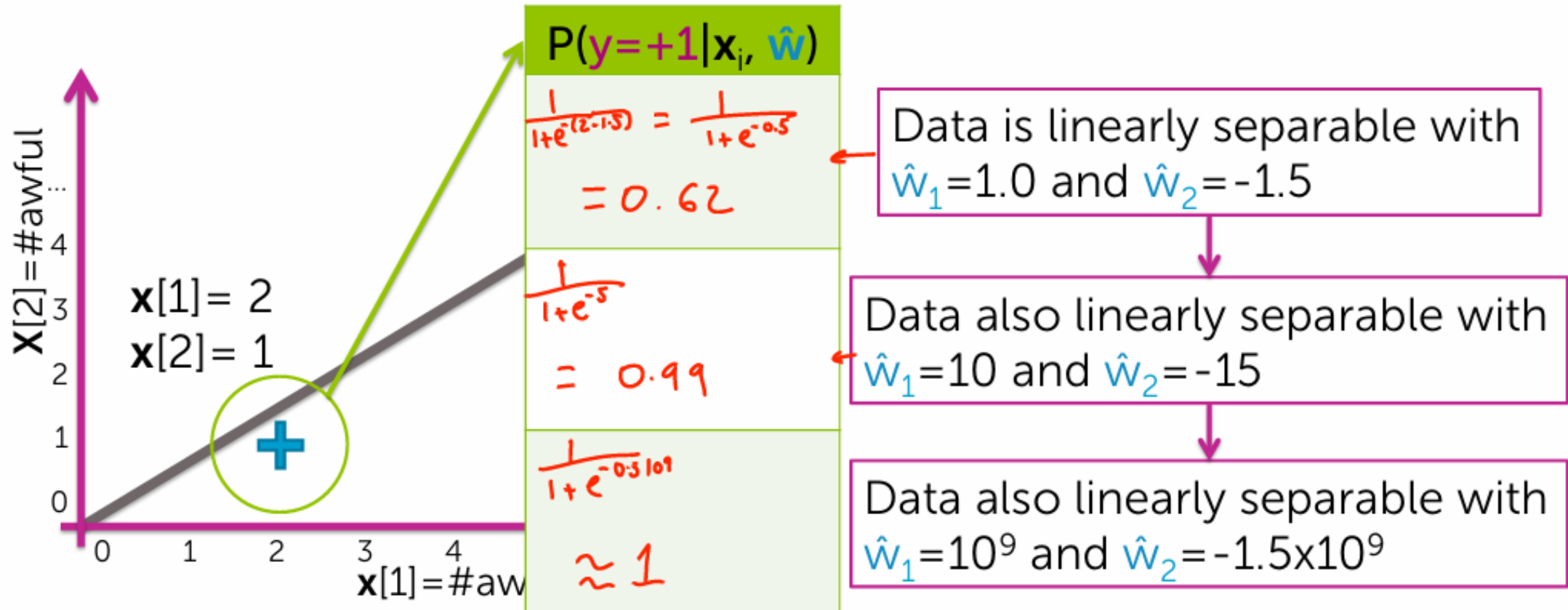
$$= 1 + 0.1 \times 1.33 = 1.133$$

Clasificadores lineales - regularización

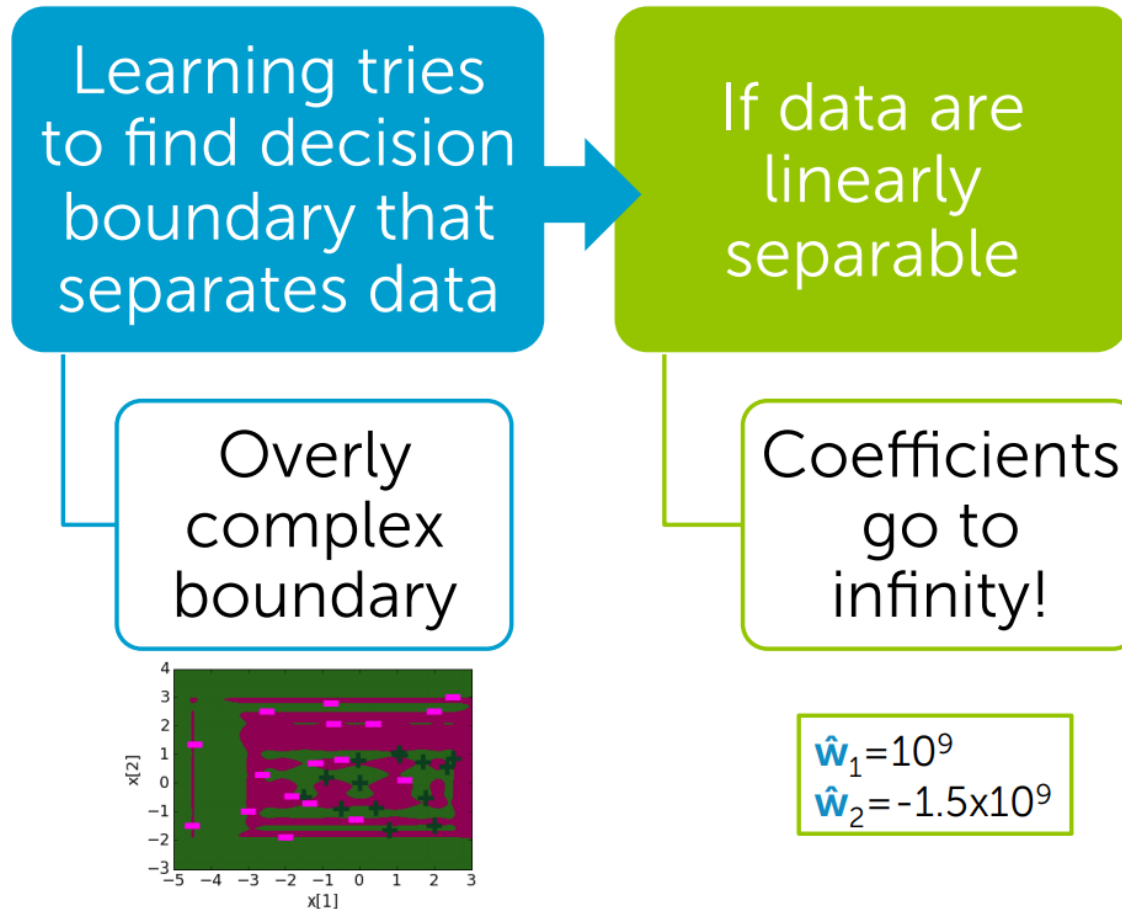
Sobreajuste



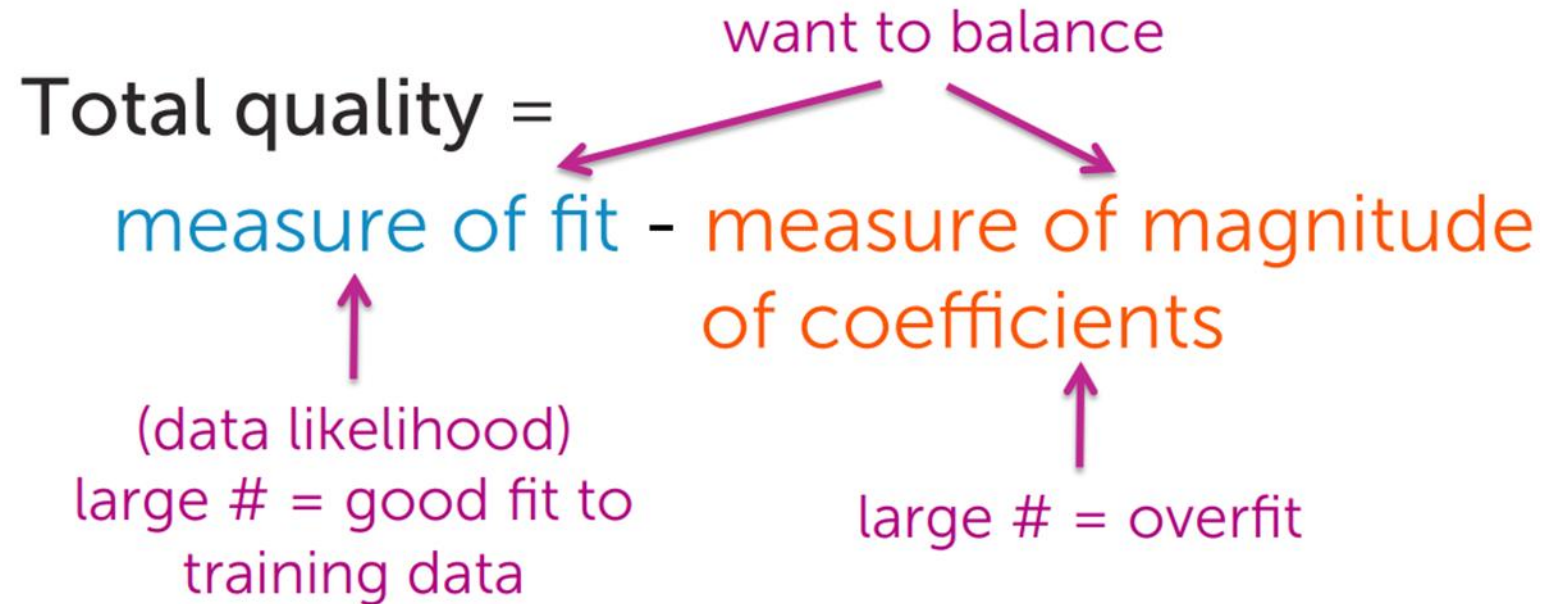
Maximum likelihood estimation (MLE)
prefers most certain model →
Coefficients go to infinity for linearly-separable data!!!



¡Maximización de verosimilitud tiende a agrandar los coeficientes!



Regularización como solución al sobreajuste



Regularización L2

What if $\hat{\mathbf{w}}$ selected to minimize

$$\ell(\mathbf{w}) - \lambda \|\mathbf{w}\|_2^2$$

 tuning parameter = balance of fit and magnitude

L_2 regularized
logistic regression

Pick λ using:

- Validation set (for large datasets)
- Cross-validation (for smaller datasets)
(see regression course)

Bias – variance tradeoff

Large λ :

high bias, low variance

(e.g., $\hat{\mathbf{w}} = 0$ for $\lambda = \infty$)

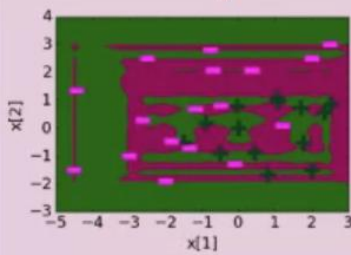
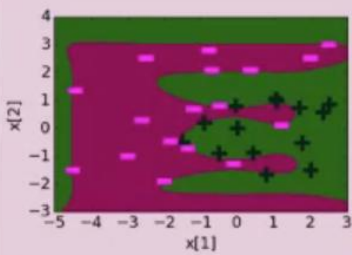
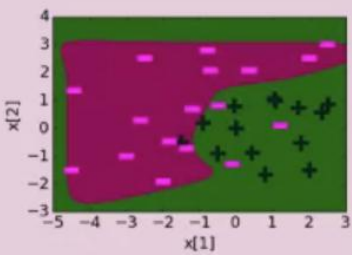
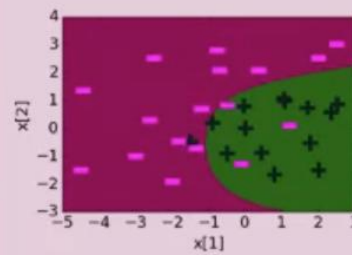
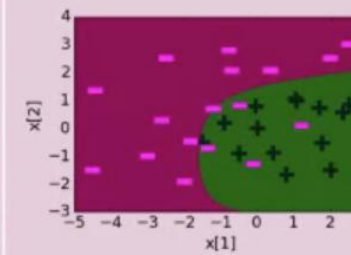
In essence, λ
controls model
complexity

Small λ :

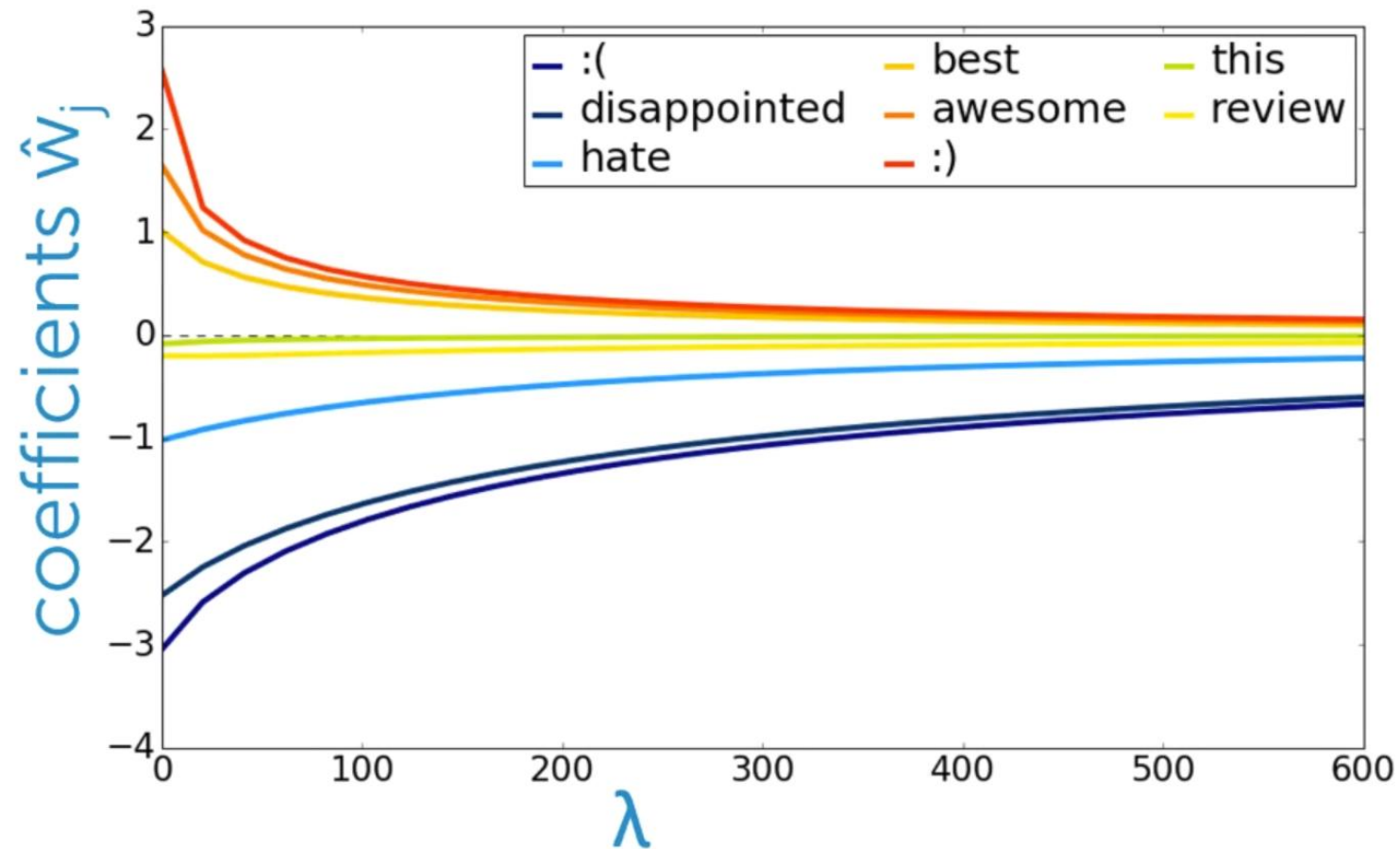
low bias, high variance

(e.g., maximum likelihood (MLE) fit of
high-order polynomial for $\lambda = 0$)

Efecto de la regularización

Regularization	$\lambda = 0$	$\lambda = 0.00001$	$\lambda = 0.001$	$\lambda = 1$	$\lambda = 10$
Range of coefficients	-3170 to 3803	-8.04 to 12.14	-0.70 to 1.25	-0.13 to 0.57	-0.05 to 0.22
Decision boundary					

Efecto de la regularización L2



Regularización L1

Total quality =

measure of fit - measure of magnitude
of coefficients

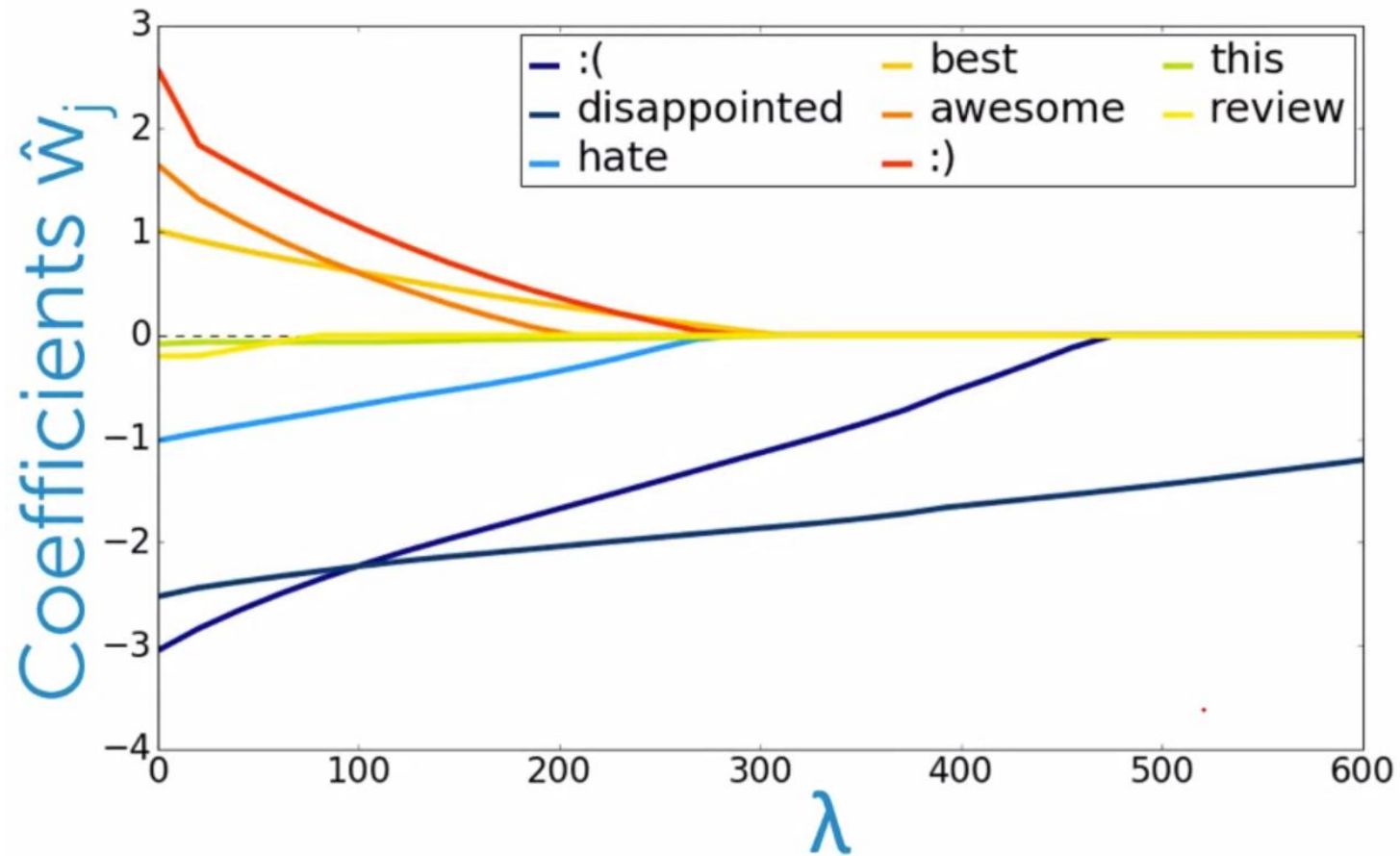
The diagram shows two horizontal curly braces. The first brace is under the text 'measure of fit' and points to the symbol $\ell(\mathbf{w})$. The second brace is under the text 'measure of magnitude of coefficients' and points to the equation $\|\mathbf{w}\|_1 = |w_0| + \dots + |w_D|$.

$$\ell(\mathbf{w}) \qquad \|\mathbf{w}\|_1 = |w_0| + \dots + |w_D|$$

L_1 regularized
logistic regression

Leads to
sparse
solutions!

Efecto de la regularización L1



Vamos al código ...